What Changes Enzyme Activity?

Biology 2e chapter 6 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

Before predicting

Catalase helps decompose hydrogen peroxide into water and oxygen. A gas syringe measures collected gas as its plunger moves. Compare equal samples while changing one variable; compare rates over the same time interval.

Investigation question

How does temperature affect the initial rate of gas collection from potato catalase?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

How to use the virtual lab

Use the assigned bath temperature. Starting the reaction starts the clock. Each trial uses fresh simulated extract and substrate. Record at 30-second intervals for 120 seconds.

Prepare the equipment in the displayed order. Observe or classify the sample, review your measurements, then click Collect data and continue. Repeat through the assigned comparison. Explore other settings after collecting the full comparison.

What Changes Enzyme Activity?

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Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: mL oxygen

Condition / sample	Repeat	Source	mL oxygen
Cold bath (10 C)	1		
Cold bath (10 C)	2		
Cold bath (10 C)	3		
Room bath (22 C)	1		
Room bath (22 C)	2		
Room bath (22 C)	3		
Warm bath (35 C)	1		
Warm bath (35 C)	2		
Warm bath (35 C)	3		

Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

What Changes Enzyme Activity?

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Copy this page for each trial. Treatment: _____ Repeat: ____ Source: _____

seconds	mL oxygen	Notes
0		
30		
60		
90		
120		

Graph your observations

Label axes and units; keep each treatment and source identifiable. For repeated measurements, connect readings only from the same sample. For ordinal scores, explain the categories instead of treating the scale as concentration.

What Changes Enzyme Activity?

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Calculate the initial collection rate using the first 30 seconds. Compare treatment means and variation.

Which setup errors could reduce measured gas without changing enzyme activity?

Does this temperature range identify an optimum or prove denaturation? Explain.

Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: Illustrative temperature response with substrate limitation; not a fit to a potato dataset. Gas dissolution, leaks, mixing delay and thermal expansion are omitted. The 10-40 C range does not model heat denaturation.